

QUIZ 2 — Feature Detection & Segmentation

Q21. Which of the following is NOT a standard step of the Canny edge detector?

Answer: D. Eigenspace projection

Q22. Non-maximum suppression in edge detection is used to:

Answer: B. Thin edges to single-pixel width

Q23. The Harris corner detector identifies corners based on:

Answer: B. Eigenvalues of the local structure (autocorrelation) matrix

Q24. In the Harris detector, a true corner region is indicated by:

Answer: C. Both eigenvalues large

Q25. SIFT stands for:

Answer: A. Scale Invariant Feature Transform

Q26. SIFT descriptors are specifically designed to be invariant to:

Answer: B. Scale and rotation

Q27. HOG stands for:

Answer: A. Histogram of Gradients

Q28. HOG features are especially well known for their use in:

Answer: B. Pedestrian/human detection

Q29. Morphological erosion applied to a binary object generally:

Answer: B. Shrinks the object

Q30. Morphological dilation applied to a binary object generally:

Answer: B. Grows the object

Q31. The morphological “opening” operation is defined as:

Answer: B. Erosion followed by dilation

Q32. The morphological “closing” operation is defined as:

Answer: B. Dilation followed by erosion

Q33. The watershed segmentation algorithm treats a grayscale image as:

Answer: B. A topographic surface to be flooded

Q34. Marker-based watershed segmentation is primarily used to reduce:

Answer: B. Over-segmentation

Q35. Region growing segmentation begins from:

Answer: B. Seed points

Q36. Region splitting and merging segmentation is commonly implemented using a:

Answer: B. Quadtree structure

Q37. In graph-based image segmentation, pixels (or regions) are typically represented as:

Answer: B. Nodes of a graph

Q38. The Normalized Cut segmentation criterion is designed to:

Answer: B. Balance segment size while minimizing the cut

Q39. Mean-shift segmentation clusters pixels by finding:

Answer: B. Modes (peaks) of the local density function

Q40. An active contour ("snake") evolves over iterations to:

Answer: B. Minimize an energy function